

Why AI Pricing Still Fails Hotels — and What Needs to Change

The architecture gap between what revenue systems promise and what they deliver

By **Dr. Tong Yin**

Founder & CEO, InsightBridge Global LLC | 20 years of senior hospitality operations | PhD, Auburn University

Most hotel executives believe they have already adopted AI in revenue management. In reality, many have adopted something closer to automated suggestion systems built on assumptions that no longer hold.

Three Broken Assumptions

Traditional revenue management systems were designed around three architectural assumptions: stable historical demand data, clearly defined competitor sets, and OTA-driven pricing signals.

In 2026, all three are increasingly invalid.

In emerging and greenfield markets — from Saudi Arabia's mega-projects to new Southeast Asian resort destinations — there is no historical baseline. In fragmented markets, competitors shift constantly. And in AI-mediated discovery environments, where a growing share of travel research now happens inside ChatGPT, Gemini, Perplexity and regional AI assistants, OTAs are no longer the sole demand signal.

Yet most pricing systems still optimise as if they were.

The result is predictable: incorrect price floors in shoulder seasons, mispriced compression nights, and excessive reliance on human override. Industry research suggests hotels deploying systems built on these outdated assumptions may leave 8–14 per cent of revenue on the table annually in markets without mature demand curves.

The Override Signal Nobody Is Capturing

The clearest evidence of this architectural mismatch is override behaviour. Across multiple markets, revenue managers routinely override system recommendations — often more than half the time — without the system learning from those decisions.

This matters because override is not noise. It is signal.

When an experienced revenue manager adjusts a price the system recommended, they are expressing tacit market knowledge — about a local event the system missed, a shift in source-market sentiment the data has not yet captured, or a competitive dynamic that defies historical pattern.

If that knowledge is captured and fed back into the system, it compounds. The system improves. The next recommendation is better. Within twelve to eighteen months, the hotel develops a pricing engine genuinely calibrated to its own market.

If that knowledge is not captured, it disappears when the revenue manager changes jobs. The system stays generic. The hotel pays for intelligence it never absorbs.

A Three-Layer Architecture

A more effective revenue approach requires structural redesign, not incremental feature upgrades.

First, demand reconstruction from first principles. Rather than forecasting solely from historical booking curves, the system ingests flight capacity data, event calendars, visa policy changes, search behaviour on AI platforms, and source-market currency movements. This is what allows pricing in markets that have no past.

Second, channel-aware net revenue optimisation. A US\$1,000 booking via OTA is not economically equivalent to a US\$1,000 direct booking once commissions, cancellation behaviour, and payment costs are netted. Revenue KPIs should reflect net contribution per available room — not gross ADR.

Third, human-in-the-loop learning systems. Every override should be treated as a training signal. The system should ask: what did the human see that I missed? This is the difference between static automation and adaptive intelligence.

The Distribution Implication

This architectural gap has a direct distribution consequence that the travel technology community has not yet fully reckoned with.

Hotels running generic pricing systems default to OTA-dependent strategies because their own intelligence is not differentiated enough to justify direct-booking investment. When the system's recommendations are no better than what the OTA's own algorithm would suggest, there is no strategic reason to build an independent distribution channel. The hotel has outsourced not just distribution, but pricing intelligence itself.

Hotels running adaptive systems — systems that learn from their own operators, absorb local market knowledge, and develop recommendations calibrated to their specific property and guest base — develop pricing confidence that supports genuine direct-booking infrastructure. Channel mix shifts not because of marketing spend, but because of intelligence advantage. The hotel knows its market better than the platform does.

As AI-mediated travel discovery reshapes the booking funnel — with a growing share of travel research moving from Google to ChatGPT, Gemini, Perplexity and regional assistants — this intelligence advantage becomes the critical differentiator. AI travel agents surface the sources they trust. Hotels with better data, better content, and more accurate pricing will be recommended ahead of intermediaries.

The future of hotel pricing will not be fully automated systems replacing humans. It will be human-amplified systems that learn faster than competitors. That is a very different category — and the hotels that understand it first will own the next decade of distribution economics.

About the author

Dr. Tong Yin is the Founder of InsightBridge Global LLC, an AI-driven hospitality intelligence and strategy advisory firm. He holds a PhD from Auburn University with research on hotel organisational performance and twenty years of senior hospitality operations experience across Asia.

tongyin@insightbridge.global · insightbridge.global